Mahesh Chitakoti

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Summary

AI & Backend Engineer with 3+ years of software engineering experience building backend services, RAG pipelines, and LLM workflows with **Python (FastAPI), TypeScript (Node.js), PostgreSQL, and Redis**. Built **LangGraph** workflows and **Qdrant** hybrid search for language learning and cut recurring LLM inference costs by **35%** through caching.

Technical Skills

Languages & Web Development: Python, TypeScript, SQL, FastAPI, Node.js, Express.js, AsyncIO, REST APIs, WebSockets, Next.js

AI & Retrieval Systems: RAG Pipelines, Agentic Workflows (LangGraph), Qdrant, pgvector, Hybrid Search (Dense + BM25), Semantic Caching, Pydantic Structured Outputs, Gemini APIs, OpenAI APIs

Databases & Cloud Infrastructure: PostgreSQL, Redis, Docker, AWS (EC2, S3), GitHub Actions CI/CD

Experience

AI & Backend Engineer (Contract) – PearlThoughts May 2026 – Present

Client Projects: Document Processing & Search *Remote*

- Engineered an automated document ingestion and data-extraction backend using **FastAPI and Redis Queues**, processing multi-page PDF invoices and client webhooks with structured schema validation.
- Developed a domain-specific RAG service using **PostgreSQL (pgvector HNSW indexing)** and semantic caching, cutting average vector retrieval latency from **280 ms to under 185 ms**.
- Containerized and deployed microservice workloads using **Docker on AWS EC2**, provisioning object storage in **S3** and establishing automated linting and test runs via GitHub Actions.

AI Developer – SahioNext Technology Pvt. Ltd. Feb 2025 – Apr 2026

Product: Tulkka, an AI-powered English learning platform *Remote*

- Designed and implemented Tulkka's lesson evaluation engine to generate structured feedback from lesson transcripts. Used **FastAPI and Gemini APIs**, with **LangGraph** for orchestration and **Pydantic** and fallback schemas to validate JSON output.
- Reduced recurring LLM inference costs by **35%** with exact-match caching and **Redis-backed semantic caching**. Mitigated upstream HTTP 429 errors with token-bucket rate limiting.
- Implemented hybrid search in **Qdrant** over **10,000+ lesson transcript chunks**, combining dense embeddings and BM25 to retrieve relevant vocabulary with p95 latency under **300 ms**.
- Developed full-duplex **WebSocket endpoints** for real-time speech evaluation, using client-side buffering and reconnection handling to recover interrupted sessions.
- Built internal dashboards in **TypeScript and Node.js** to curate datasets and combine reviewer rubric scores with automated model feedback.

Full Stack Developer – The DevOps Team Oct 2023 – Jan 2025

Product: Recruiting Platform *Bengaluru, India*

- Profiled bottleneck queries and optimized **PostgreSQL indexing** alongside **Redis caching**, reducing median API response times from **800 ms to 320 ms (60% decrease)**.
- Optimized candidate search queries with multicolumn indexes to reduce database query latency.
- Developed multi-step candidate onboarding forms in **Next.js** and A/B tested changes that increased registration conversion by **18%**.

Projects

WorkBridge – Project Collaboration Platform *Sep 2026 – Present*

Python, FastAPI, TypeScript, Node.js, Express.js, PostgreSQL, pgvector, Next.js | Live Project

- Created a platform where founders and hiring teams evaluate developers through project submissions linked to **GitHub repositories**.
- Engineered a **FastAPI** service using **LLMs and Pydantic structured outputs** to analyze submitted GitHub repositories for code quality, architecture patterns, and task completion.
- Implemented **PostgreSQL (pgvector)** semantic search to match founders with developers by verified project history, backed by a **Node.js/Express API gateway** for GitHub OAuth 2.0 and RBAC.

Certifications

Oracle Agentic AI Certified Foundations Associate – Oracle — Aug 2026
Oracle AI Vector Search Certified Professional – Oracle — Sep 2025
Machine Learning I – Columbia University — Aug 2025
Oracle Cloud Infrastructure Generative AI Professional – Oracle — Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU) — Chikkamagaluru, KA, India
Bachelor of Engineering (B.E.) — *Aug 2023*